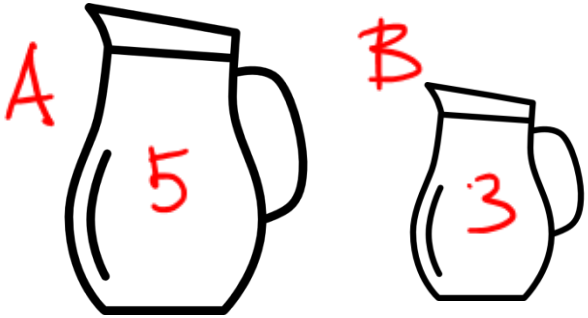
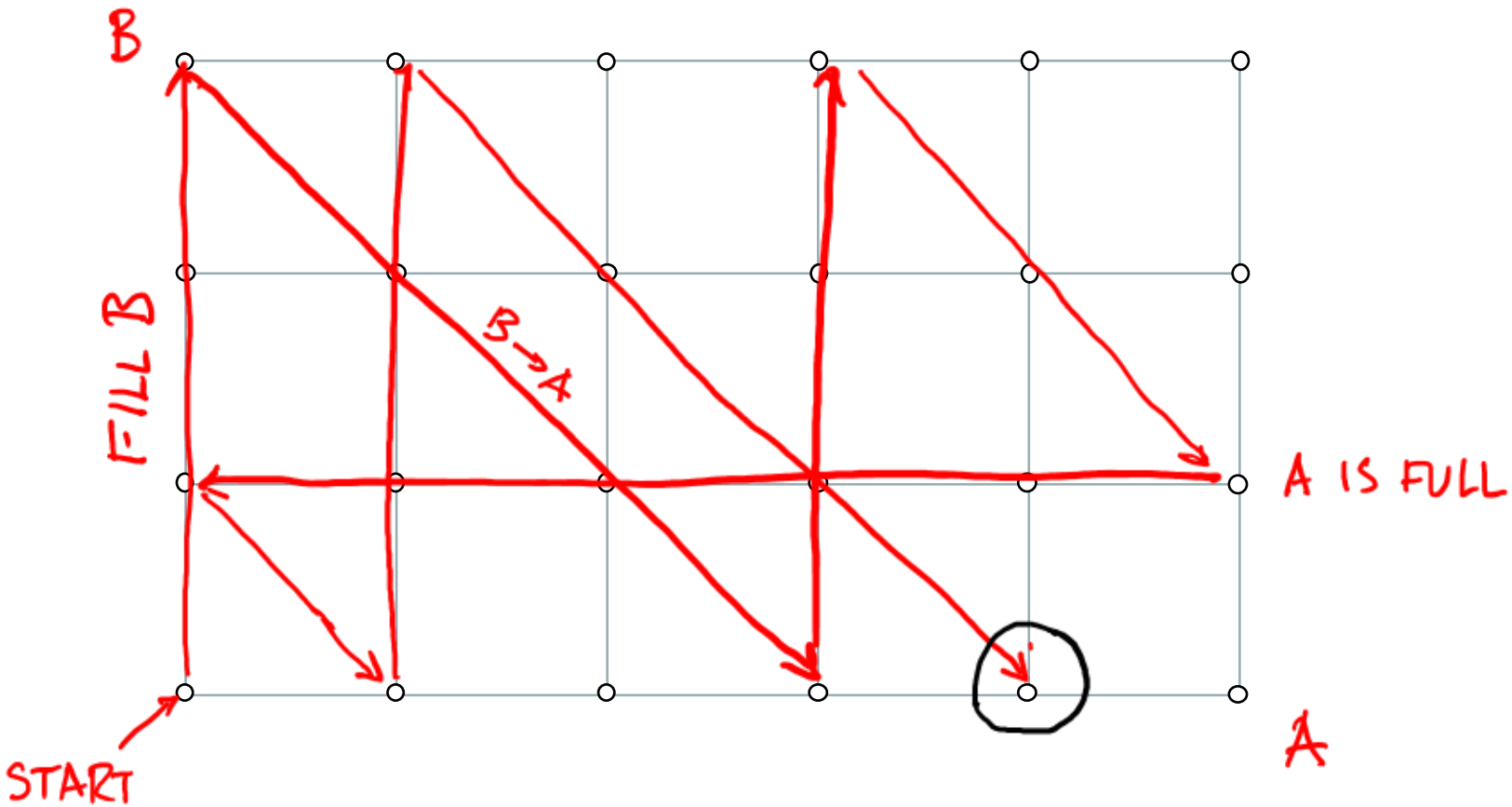
CSI 2101: Discrete Structures

4. NUMBERS

Andrej Bogdanov



FILL B	–	3
B → A	3	–
FILL B	3	3
B → A	5	1
SPILL A	–	1
B → A	1	–
FILL B	1	3
B → A	4	–


B
FILL B
B→A
A IS FULL
START
A

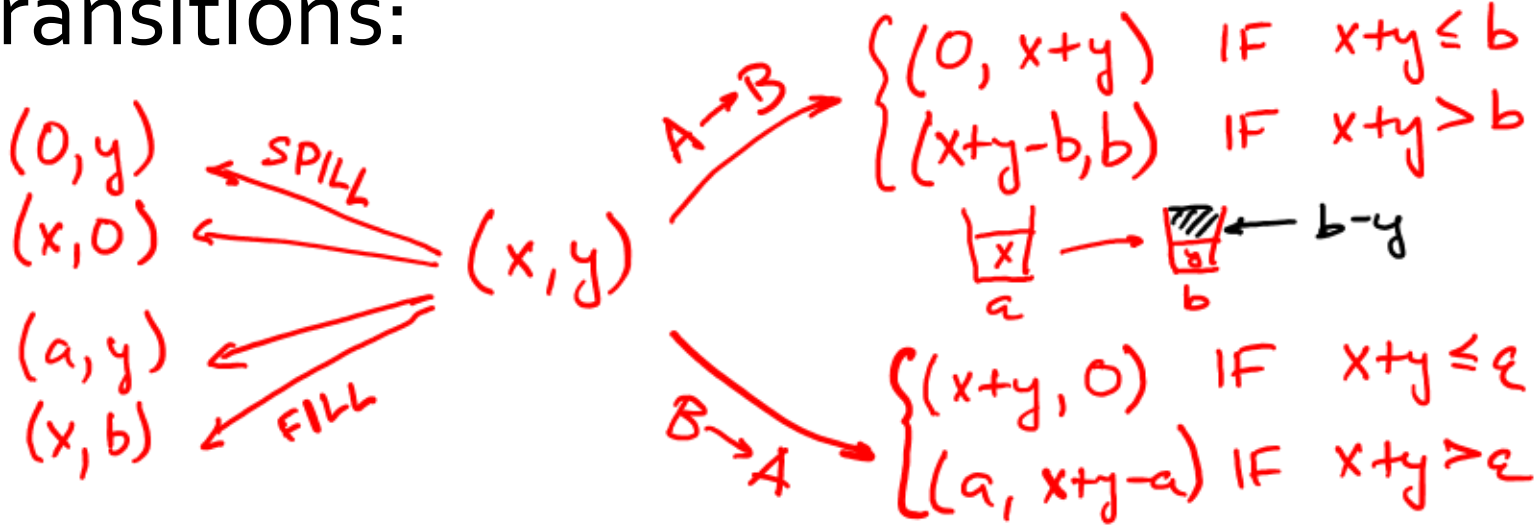
$73l, 50l \rightarrow 4l$?

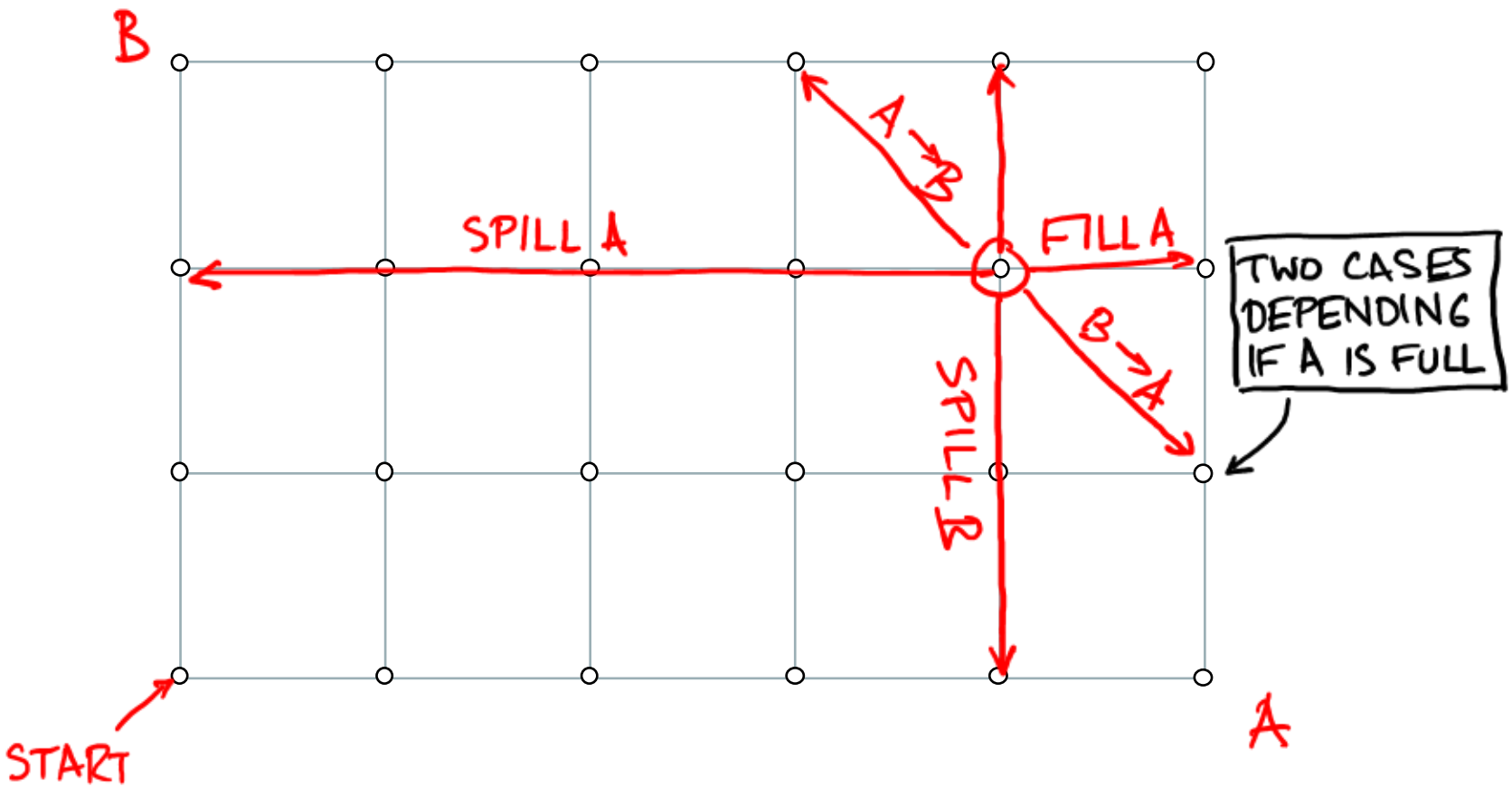
$9l, 6l \rightarrow 4l$?

$1147l, 899l \rightarrow 999l$?

states: $(x,y) : 0 \le x \le a,\ 0 \le y \le b$

transitions:




B
A
SPILL A
FILL A
A → B
B → A
SPILL B
TWO CASES DEPENDING IF A IS FULL
START

integer linear combinations

A <u>COMBINATION</u> OF a AND b IS ANY NUMBER OF THE FORM $s \cdot a + t \cdot b$

<u>Ex.</u>

$$\left.\begin{aligned} 6 &= 3 \cdot 12 - 2 \cdot 15 \\ 9 &= 2 \cdot 12 - 15 \end{aligned}\right\}$$ 6 AND 9 ARE BOTH COMBINATIONS OF 12 AND 15

$$0 = 0 \cdot a + 0 \cdot b$$ } 0 IS A COMBO OF ANY a, b

Invariant: x AND y (WATER AMOUNT IN JUGS) ARE COMBINATIONS OF a AND b

<u>Proof.</u> TRUE FOR START STATE; ALL TRANSITIONS SET x' AND y' TO SOME COMBINATION OF x, y, a, b.

Lemma: Combinations of combinations of a and b are combinations of a and b

$$6 = 3 \cdot 12 - 2 \cdot 15$$
$$9 = 2 \cdot 12 - 15$$

$$3 = -1 \cdot 6 + 9$$
$$= -1(3 \cdot 12 - 2 \cdot 15) + (2 \cdot 12 - 15)$$
$$= (-1 \cdot 3 + 2) \cdot 12 + (2-1) \cdot 15$$
$$= -1 \cdot 12 + 15$$

Proof. ASSUME

$$x = sa + tb$$
$$y = ua + vb$$

$$px + qy =$$
$$= p(sa + tb) + q(ua + vb)$$
$$= (ps + qu)a + (pt + qv)b \quad \square$$

divisibility

$d|a$ (d DIVIDES a) IF $a = kd$ FOR SOME k

$2|4$ $\quad 3|6$ $\quad 2\nmid 5$ $\quad 4\nmid 2$ $\quad$ ANYTHING $|0$

$4 = 2\cdot 2$ $\quad 6 = 3\cdot 2$ $\quad 0 = 0\cdot$ ANYTHING

Lemma: If $d|a$ and $d|b$ then d divides all combinations of a and b.

<u>Proof.</u> $a = pd,\ b = qd \rightarrow sa + tb = (sp + tq)d.$ $\square$

Corollary: Bruce & Sam DO NOT SURVIVE $a=9, b=6, v=4$.

<u>Proof.</u> $3|6,\ 3|9$. BY Invariant 3 MUST ALWAYS DIVIDE WATER A OUNT. BUT $3\nmid 4$. $\square$

invariant recap

Water in each jug is combination of a and b

$d|a,\ d|b \rightarrow d|$ ALL MEASURABLE AMOUNTS

Can use it to show impossibility

$3|9,\ 3|6$ BUT $3\nmid 4 \rightarrow 4$ IS NOT MEASURABLE

Does not tell us what is possible

$a=26,\ b=21 \xrightarrow{?} v=4$

Die Hard: the algorithm

REPEAT UNTIL ℓ IN JUG A:

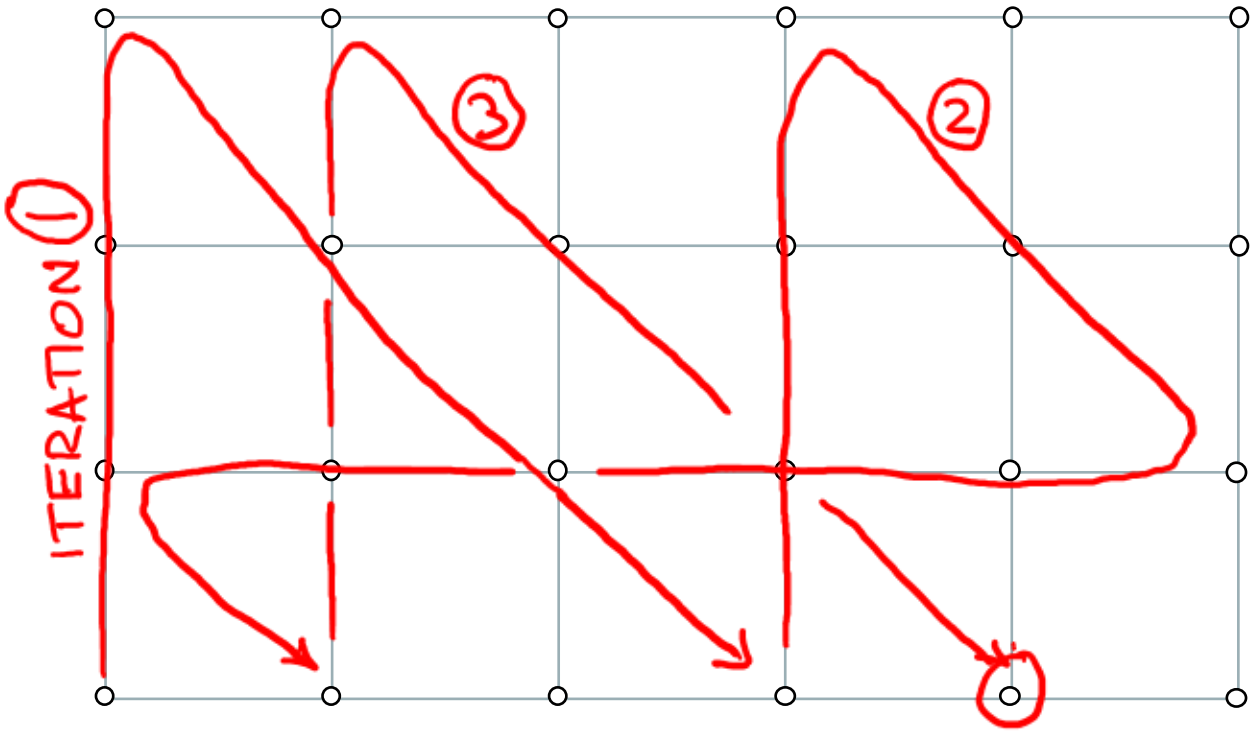
ITERATION

- FILL JUG B
- B → A
- IF JUG A IS FULL
  - SPILL JUG A
  - B → A

A	
+b	IF B NOT FULL
−a	
−a+b	IF B FULL

Lemma: AFTER s ITERATIONS, JUG A HAS $sb - ?a$ LITRES FOR SOME ?.

Proof. INITIALLY $0 = 0b - 0a$. IN EACH ITERATION A CHANGES BY $+b$ OR $+b-a$ LITRES. □



JUG A HAS 3.3 - 1.5 LITRES

ITERATIONS

SPILLS

Lemma: If v is a combination of a and b then Bruce and Sam survive.

Ex. $a=5 \quad b=3 \quad v=4$

$$4 = -2\cdot\underline{3} + 2\cdot\underline{5}$$

WANT POSITIVE

$$+\ 0 = 5\cdot\underline{3} - 3\cdot\underline{5}$$

$$4 = 3\cdot\underline{3} - 1\cdot\underline{5}$$

AFTER $s=3$ ITERATIONS JUG A HAS $3\cdot\underline{3} - ?\cdot 5\,\ell$

? MUST BE 1

?	0	1	2
$3\cdot\underline{3} - ?\cdot 5$	9	4	-1
	OVERFLOW		UNDERFLOW

Proof. $v = s\cdot\underline{b} + t\cdot\underline{a}$

IF s IS NEGATIVE KEEP ADDING $a\cdot\underline{b} - b\cdot\underline{a}$ TILL POSITIVE

BY Invariant AFTER s ITERATIONS MUST HAVE $s\cdot\underline{b} - ?\cdot\underline{a}$ LITRES IN JUG A.

? MUST EQUAL t:

$$0 \le s\underline{b} - t\underline{a} < \underline{a}$$

$$sb - (t+1)\underline{a} < 0 \quad \text{UNDERFLOW}$$

$$sb - (t-1)\underline{a} > \underline{a} \quad \text{OVERFLOW}$$

$73l, 50l \longrightarrow 4l$?

~~$9l, 6l \longrightarrow 4l$?~~

$1147l, 899l \longrightarrow 999l$?

Important Theorem:

$\mathbf{gcd}(a, b)$ is a combination of a and b.

$$1 = \gcd(73, 50) = 19 \cdot \underline{50} - 13 \cdot \underline{73}$$

division with remainder

Theorem: For all n, $d > 0$ there exist unique q, $0 \leq r < d$ such that $n = qd + r$.

Existence

LET q = LARGEST MULT. OF d FOR WHICH $qd \leq n$

$r = n - qd \geq 0$

$(q+1)d > n \rightarrow n - qd < d.$

Uniqueness:

SUPPOSE

$$n = qd + r$$
$$n = q'd + r'$$
$$0 = (q-q')d + (r-r')$$
$$(q-q')d = \underbrace{r'-r}_{<d \text{ IN ABS VALUE}}$$

MUST $=0$

gcd and Euclid's algorithm

$\boxed{n \geq d \geq 0}$

$E(n,d)$: IF $d=0$ RETURN n
ELSE RETURN $E(d,r)$ $(n = qd + r)$.

$$
\begin{aligned}
E(73, 50) &= E(50, 23) \\
&= E(23, 4) \\
&= E(4, 3) \\
&= E(3, 1) \\
&= E(1, 0) \\
&= 1
\end{aligned}
$$

$$
\begin{aligned}
73 &= 50 + 23 \\
50 &= 2 \cdot 23 + 4 \\
23 &= 5 \cdot 4 + 3 \\
4 &= 1 \cdot 3 + 1 \\
3 &= 3 \cdot 1 + 0
\end{aligned}
$$

$$E(1147, 899) = E(899, 248)$$
$$= E(248, 155)$$
$$= E(155, 93)$$
$$= E(93, 62)$$
$$= E(62, 31)$$
$$= E(31, 0)$$
$$= 31$$

$1147 = 899 + 248$

$899 = 3 \cdot 248 + 155$

$248 = 155 + 93$

$155 = 93 + 62$

$93 = 62 + 31$

$62 = 2 \cdot 31$

Theorem: Euclid's algorithm always terminates and outputs the GCD.

<u>TRANSITIONS</u>: $(n,d) \to (d,r)$

<u>Invariant</u>: GCD STAYS THE SAME.

<u>Proof</u>. $\to$: ANY $t \mid n,d \to t \mid d,r$

$r = \underline{n} - q \cdot \underline{d}$ = COMBO OF n,d

$\leftarrow$ ANY $t \mid dr \to t \mid n,d$

$n = q\underline{d} + \underline{r}$ = COMBO OF d,r. $\square$

extended Euclid's algorithm

$$E(73, 50) = E(50, 23)$$
$$= E(23, 4)$$
$$= E(4, 3)$$
$$= E(1, 0)$$

$$73 = 50 + 23 \quad (6)$$
$$50 = 2 \cdot 23 + 4 \quad (4)$$
$$23 = 5 \cdot 4 + 3 \quad (2)$$
$$4 = 1 \cdot 3 + 1 \quad (1)$$

$$1 = 4 - 3 \quad (3)$$

$$3 = 23 - 5 \cdot 4$$
$$4 = 50 - 2 \cdot 23$$
$$23 = 73 - 50$$

$$1 = -23 + 6 \cdot 4 \quad (5)$$
$$1 = 6 \cdot 50 - 13 \cdot 23 \quad (7)$$
$$1 = 19 \cdot 50 - 13 \cdot 73$$

Theorem: Extended Euclid outputs s and t that satisfy $sa + tb = \mathbf{gcd}(a, b)$

HOW TO WRITE $v = s'a + t'b$?

RECALL $\gcd(a,b) \mid v$ SO

$$v = k \cdot \gcd = ks \cdot a + kt \cdot b$$

Ex. $4 = ? \cdot 73 + ? \cdot 50$

$1 = 13 \cdot 73 - 19 \cdot 50$ SO

$4 = 52 \cdot 73 - 76 \cdot 50.$

Corollary: Bruce and Sam survive if and only if v is a multiple of $\mathbf{gcd}(a, b)$.

Proof. $\longleftarrow$ IMPORTANT THEOREM

$\gcd(a,b) = s\underline{a} + t\underline{b}$

$v = k \cdot \gcd$ FOR SOME k

$v = (ks)\underline{a} + (kt)\underline{b}$ COMB. OF $\underline{a}$ AND $\underline{b}$

$\longrightarrow$ CONTRAPOSITIVE

$\gcd(a,b) \mid a, b$ BUT $\gcd(a,b) \nmid v \rightarrow$ DIE

prime factorization

Every number is a product of primes.

$$36 = \begin{cases} 2 \cdot 18 = 2 \cdot 2 \cdot 9 = 2 \cdot 2 \cdot 3 \cdot 3 \\ 3 \cdot 12 = 3 \cdot 2 \cdot 6 = 3 \cdot 2 \cdot 2 \cdot 3 \end{cases}$$

SAME RESULT!

Fundamental Theorem of Arithmetic:

The prime factorization is unique.

$$2 \cdot 3 \cdot 7 \stackrel{?}{=} 3 \cdot 3 \cdot 5$$

$2 \mid$ LHS BUT $2 \nmid$ RHS

$2 \nmid 3 \cdot 3 \cdot 5 \longleftarrow 2 \nmid 3$ AND $2 \nmid 3$ AND $2 \nmid 5$

Lemma: If prime $\overset{P}{p \mid ab}$ then $\overset{A}{p \mid a}$ or $\overset{B}{p \mid b}$.

FALSE IF p COMPOSITE: $4 \mid 2 \cdot 2$ BUT $4 \nmid 2$, $4 \nmid 2$.

PROOF STRATEGY: $\dfrac{P \text{ AND } (\text{NOT } A) \rightarrow B}{P \rightarrow A \text{ OR } B}$

<u>Proof.</u> ASSUME $p \mid ab$ BUT $p \nmid a$.

$\downarrow$

$1 = sp + ta \leftarrow \gcd(p,a) = 1$

FOR SOME s, t

$b = sbp + tab$

$p \mid p$ AND $p \mid ab \rightarrow p \mid b$. $\square$

gcd and factorization

SCHOOLBOOK GCD

$18 = 2 \cdot 3 \cdot 3$

$12 = 2 \cdot 2 \cdot 3$

$\gcd = 2 \cdot 3 = 6$

REQUIRES <u>FACTORING</u>

EUCLID'S

$E(18,12) = E(12,6)$ $\quad 18 = 12 + 6$

$= E(6,0)$ $\quad 12 = 2 \cdot 6 + 0$

$= 6$